RSYNC DOCUMENTATION

RSYNC: THE ULTIMATE SSD-SAFE BACKUP GUIDE

A technical manual for efficient, integrity-focused vault syncing with minimal storage wear.

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1. Introduction

`rsync` (Remote Sync) is a powerful utility for local and remote file synchronization. This guide focuses on the "SSD-Saver" methodology—prioritizing data integrity while minimizing the unnecessary write operations that shorten the lifespan of Solid State Drives (SSDs).

Why "SSD-Saver"?

Standard sync tools often create temporary copies of files before moving them to the destination. On a large dataset, this doubles the total bytes written. By using the `--inplace` flag, `rsync` patches files directly, significantly reducing write amplification.

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2. Core Backup Workflows

These commands are optimized for 99% of synchronization scenarios.

2.1 The Daily Sync (Performance Focus)

Use this for regular backups. It only writes modified blocks, saving massive wear and time.

```
sudo rsync -avhP --delete --inplace /source/path/ /destination/path/
```

2.2 The Paranoid Sync (Integrity Focus)

Use this when you need absolute certainty that every bit is identical (e.g., after a system crash or for cold storage). It reads both sides entirely but only writes if differences are found.

```
        sudo rsync -avhP --delete --inplace --checksum /source/path/
/destination/path/
```

2.3 The Safety First (Dry Run)

Always run this first. It simulates the process and lists exactly what would be changed or deleted without touching a single byte.

```
        sudo rsync -avhPn --delete --inplace /source/path/ /destination/path/
```

2.4 Remote Syncing via SSH

`rsync` natively supports SSH for secure transfers between different machines. By default, it uses the SSH protocol if a remote host is specified in the destination or source.

Remote Mirror Example:

```
        sudo rsync -avhPn --delete --inplace /media/veracrypt1/
192.168.13.1:/media/veracrypt2
```

Note: Ensure SSH keys are configured for seamless automation.

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3. Flag Reference Guide

FLAG	NAME	TECHNICAL IMPACT
`-a`, `--archive`	**Archive**	Preserves permissions, owners, symlinks, and timestamps. Essential for "mirrors."
`-v`, `--verbose`	**Verbose**	Provides real-time feedback on which files are being processed.
`-h`, `--human-readable`	**Human-Readable**	Converts byte counts into intuitive units (GB, MB, KB).
`-P`	**Partial/Progress**	Shows a progress bar and allows resuming of interrupted large transfers. (Equivalent to `--partial --progress`)
`--delete`	**Delete**	Removes files in the destination that no longer exist in the source.
`--inplace`	**Direct Patch**	**CRITICAL:** Updates files in place. Prevents temporary file creation and reduces SSD wear.
`-c`, `--checksum`	**Checksum**	Forces bit-by-bit comparison instead of relying on file size and modification time.
`-i`, `--itemize-changes`	**Itemize Changes**	Outputs a change-summary for all updates, providing granular detail on *why* a file is being synced (e.g., size, time, or permissions).
`-n`, `--dry-run`	**Dry Run**	Simulates the operation. Prevents catastrophic accidental deletions.

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4. Pro-Tips & Common Pitfalls

The Trailing Slash Rule

The presence or absence of a trailing slash on the **source** path changes everything: -

`/source/` (With slash): Copies the **contents** of the folder. - `/source` (No slash): Copies the **folder itself** into the destination.

Permission Handling

Running with `sudo` is often necessary to preserve file ownership and special permissions across different user directories.

Verification

After a critical sync, you can run the "Paranoid Sync" command with the `--dry-run` flag. If it reports 0 files changed, your backup is a bit-perfect mirror.

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